

AI Disclosure Appendix

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DA25D400

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1 The Tool Used

ChatGPT was used.

2 The Specific Prompts Provided

- 1) Should `config.yaml` have all the hardcoded file links? Also, should I update the version directly from the code or manually after a git commit?
- 2) Is a git commit necessary after each data version update?
- 3) Will the `manifest.txt` file be manually or automatically appended (to store which script produced which data version)?
- 4) What is the equivalent command for `mkdir` for a string object in Windows?
- 5) I created a `data_prep.py` file. Later, I modified it. Should the previous file be completely replaced with the new one, or should it be modified incrementally?
- 6) The following warning occurred:

```
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row_indexer, col_indexer] = value instead.
```

```
X_train[numeric_cols] = scaler.fit_transform(X_train[numeric_cols])
```

```
c:\Users\karan\OneDrive\Desktop\KaranKishore\Academics\  
PhD_Sem2\MLOps\Assignment_1>manual_mlops_project\  
src\data_prep.py:41: SettingWithCopyWarning
```

Reference documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

Fix the above error.

- 7) The web interface only prints column values. There are no input fields or prediction interface. How can this be considered production-grade? There is no UI and no example values. Can statistics such as mean values be added to help provide reasonable inputs?
- 8) Inputs cannot be provided. There appears to be an issue in the displayed interface.
- 9) Regardless of input, the output remains constant:

```
{
  "Torque_Nm": 4,
  "Tool_wear_min": 150,
  "TWF": 0,
  "HDF": 0,
  "PWF": 0,
  "OSF": 0,
  "RNF": 0,
  "Type_L": 1,
  "Type_M": 0
}

model_version: v3
prediction: 0
probability_class_0: 0.615
probability_class_1: 0.385
```

The probabilities (0.615 and 0.385) remain constant. What could be the possible reason for this stagnant output?

- 10) The following test script was written:

```
import requests

BASE_URL = "http://127.0.0.1:8000"

def test_health():
    r = requests.get(f"{BASE_URL}/")
    assert r.status_code == 200
    data = r.json()
    assert "model_version" in data

def test_schema():
    r = requests.get(f"{BASE_URL}/schema")
    assert r.status_code == 200
    data = r.json()
```

```

        assert "features" in data

def test_prediction():
    payload = {
        "Air temperature [K]": 298.9,
        "Process temperature [K]": 309.1,
        "Rotational speed [rpm]": 2861,
        "Torque [Nm]": 4.6,
        "Tool wear [min]": 143,
        "TWF": 0,
        "HDF": 0,
        "PWF": 1,
        "OSF": 0,
        "RNF": 0,
        "Type_L": 1,
        "Type_M": 0
    }

    r = requests.post(f"{BASE_URL}/predict", json=payload)
    assert r.status_code == 200
    data = r.json()
    assert "prediction" in data

def test_schema_valid():
    payload = {"random_attribute": 123}
    r = requests.post(f"{BASE_URL}/predict", json=payload)
    assert r.status_code in [400, 422]

if __name__ == "__main__":
    test_health()
    test_schema()
    test_prediction()
    test_schema_valid()
    print("All smoke tests passed")

```

Is this code correct? A 422 error is still displayed. Where could the error have originated?

11) The following error occurred:

```

File "...", line 34, in <module>
    pred_log["prediction"] == day2_df[target_col]

KeyError: 'prediction'

```

12) Give me a boilerplate to run fastAPI endpoint.

- 13) How do I display an example input in the web interface? Also 127.0.0.1:8000 doesn't display the options like POST and PREDICT, where do I find them?
- 14) How to setup a task scheduler in Windows (similar to cronjob in Linux)?

3 The sections of the assignment where

A few examples of inputs for smoke tests centered around the mean values. Bollerplate for fastAPI endpoint. Also, took help in understanding task scheduler in Windows.